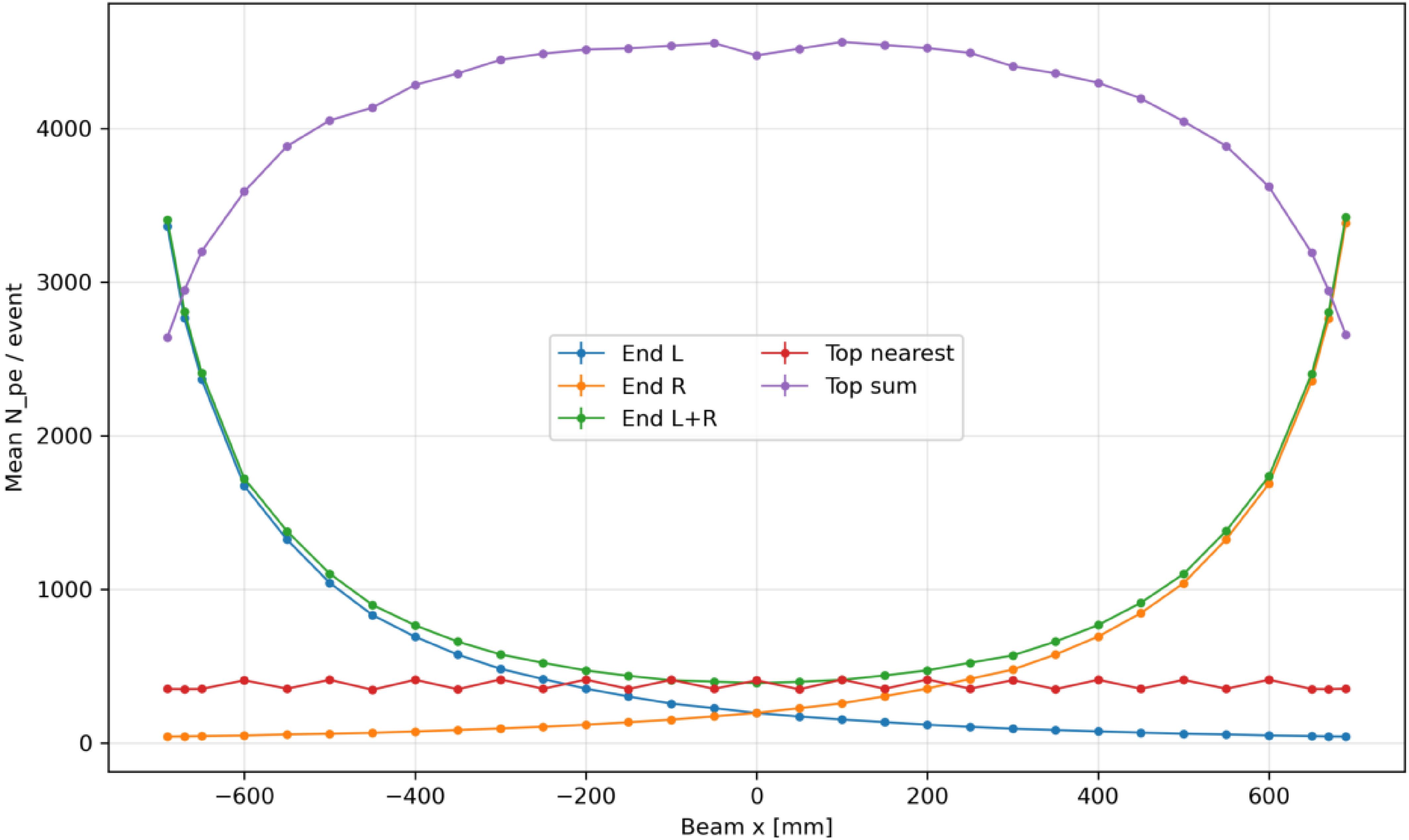
EXEC_08 - EXEC_07 EndTop photon budget

31 positions x 2000 events, 86 channels, EJ-204 OPSC-101, Broadcom AFBR-S4N66P024M

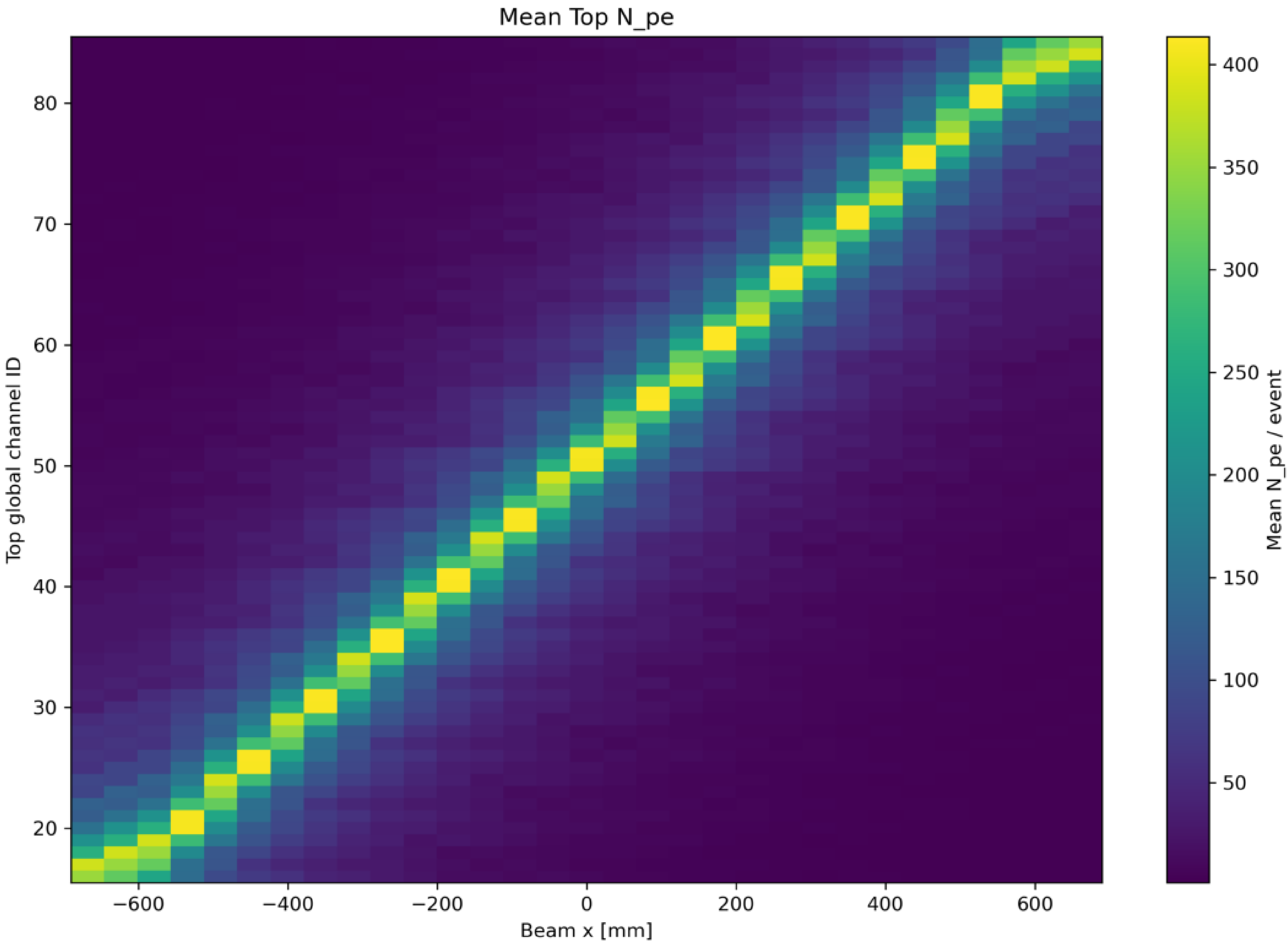
Intrinsic simulation: /sipm/jitterSigma 0

- Effective attenuation: left 33.14 +/- 0.03 cm; right 33.06 +/- 0.03 cm. These include light extraction through 70 Top windows and are not expected to equal t
- Apparent slope-derived velocity from mean all-photon time: left 27.68 +/- 0.27 cm/ns; right 27.67 +/- 0.27 cm/ns. EXEC_10 shows this is not a direct propag
- Typical nearest-Top var/mean: median 24.81, range 19.14-29.27. N_pe is not forced to Poisson.
- Intrinsic End SUM4 sigma_group= sigma(DeltaT)/sqrt(2): mean 148.28 ps, range 12.10-437.32 ps. No SPTR/electronics jitter.
- sigma(t_avg) mean 142.49 ps, range 93.20-225.36 ps.

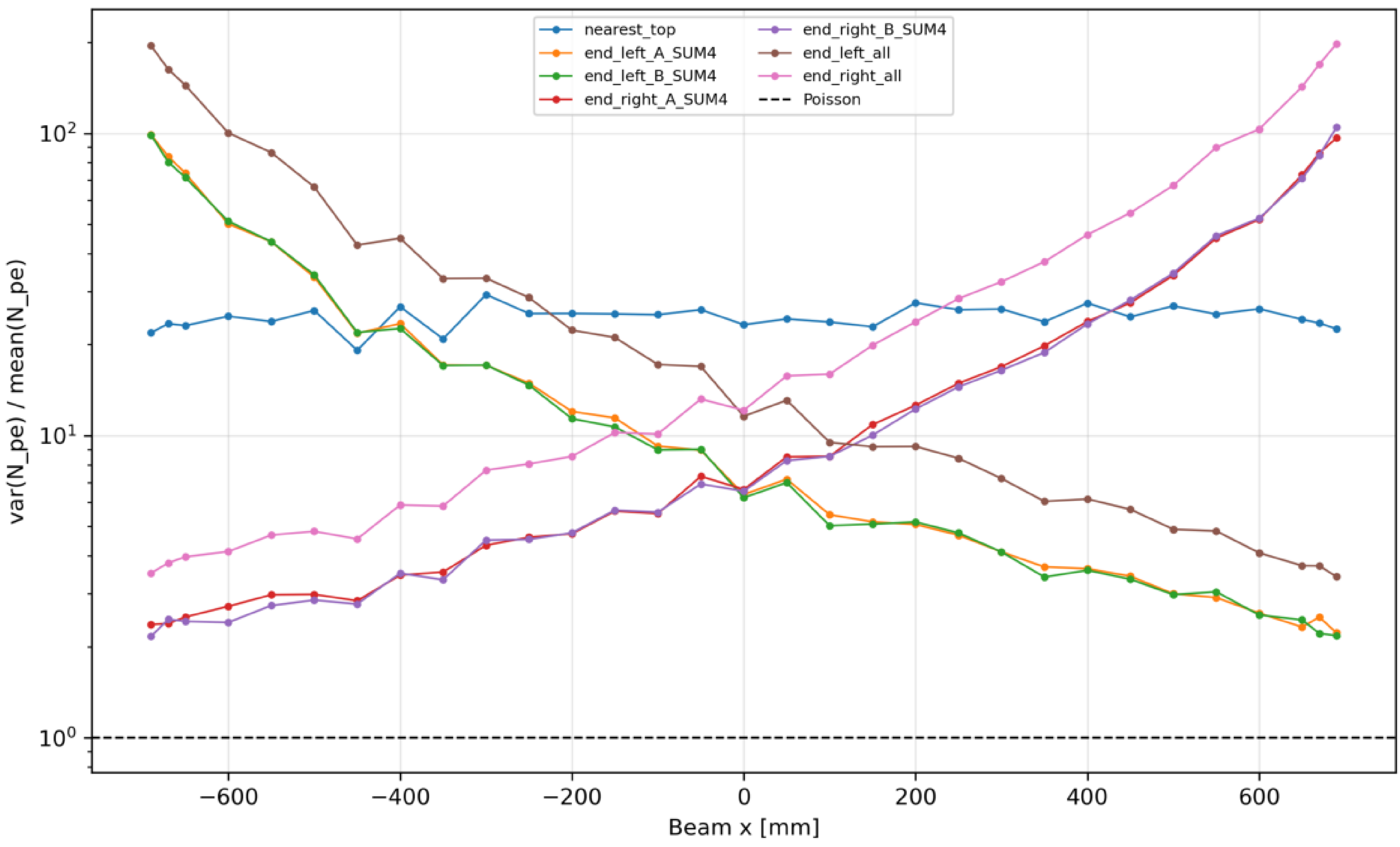
Top uses 70 simulated SiPM elements and does not replicate the 32-SiPM hardware setup.



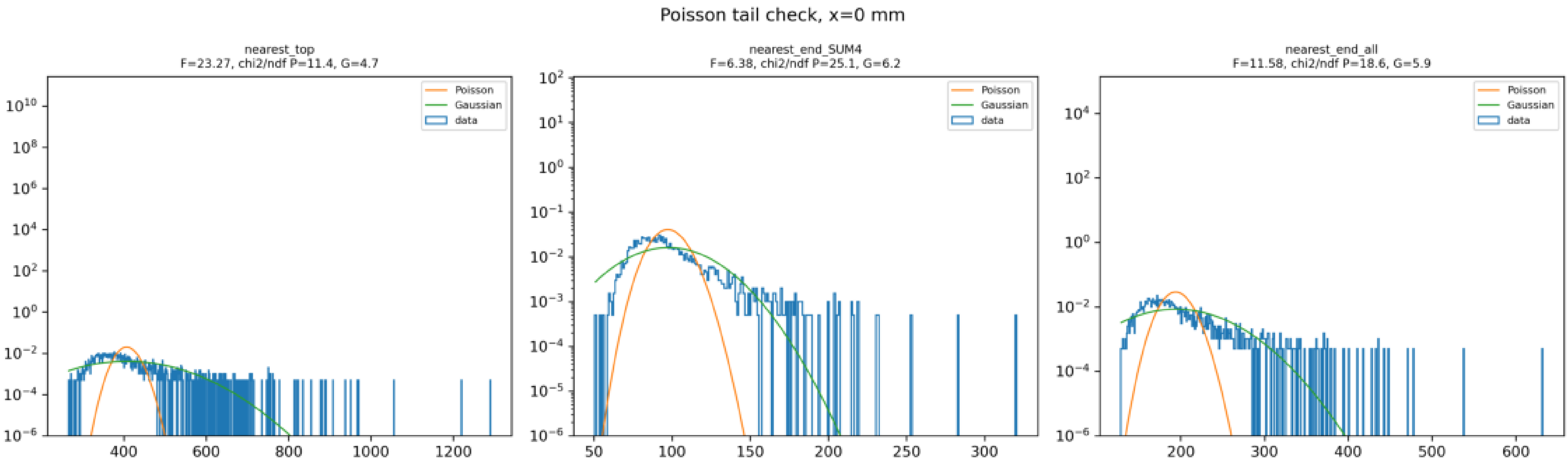
End L
End R
End L+R
Top nearest
Top sum
Mean N_pe / event
Beam x [mm]
0
1000
2000
3000
4000
−600
−400
−200
0
200
400
600

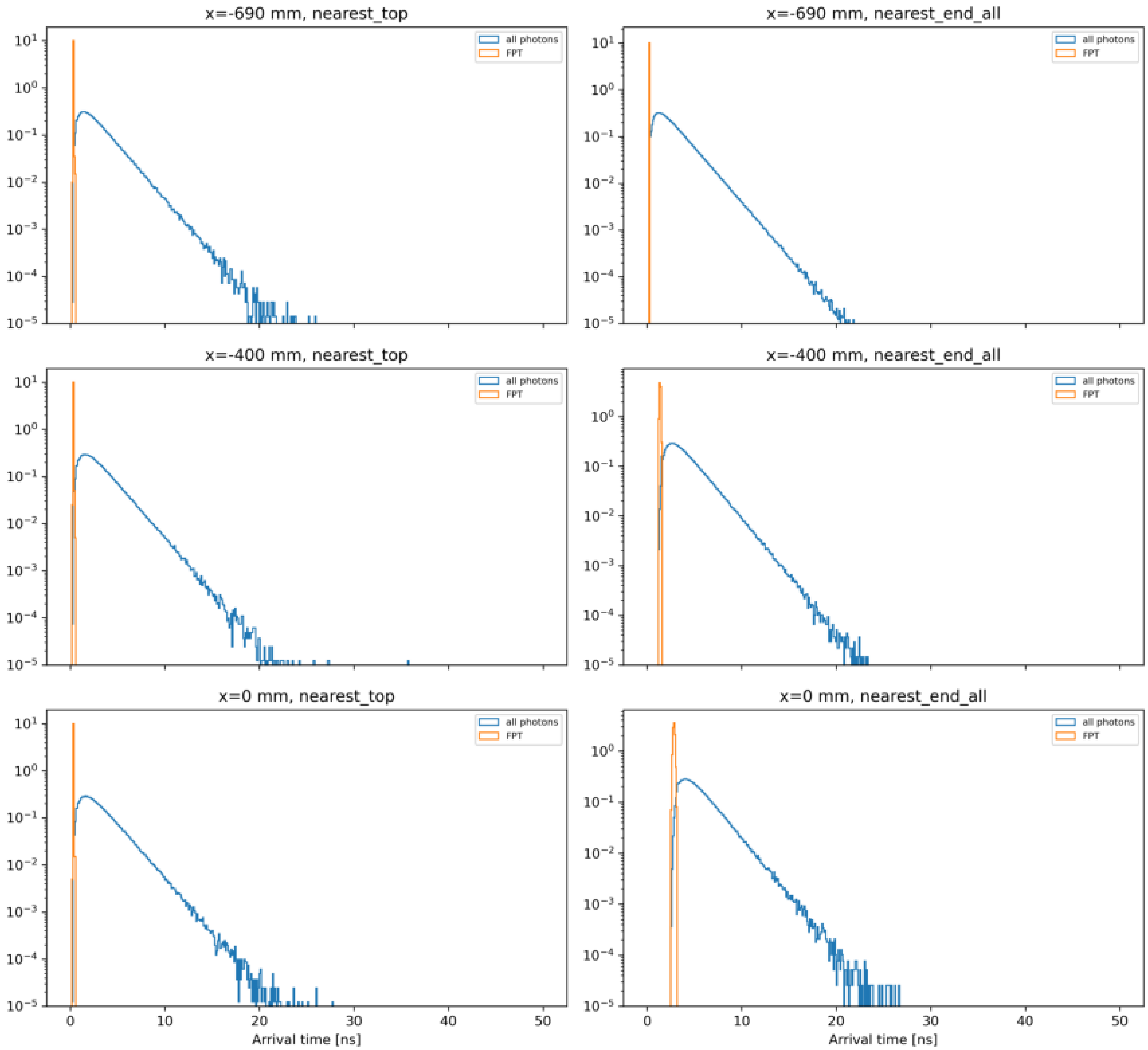


Mean Top N_pe
Top global channel ID
Beam x [mm]
Mean N_pe / event
20
30
40
50
60
70
80
−600
−400
−200
0
200
400
600
50
100
150
200
250
300
350
400

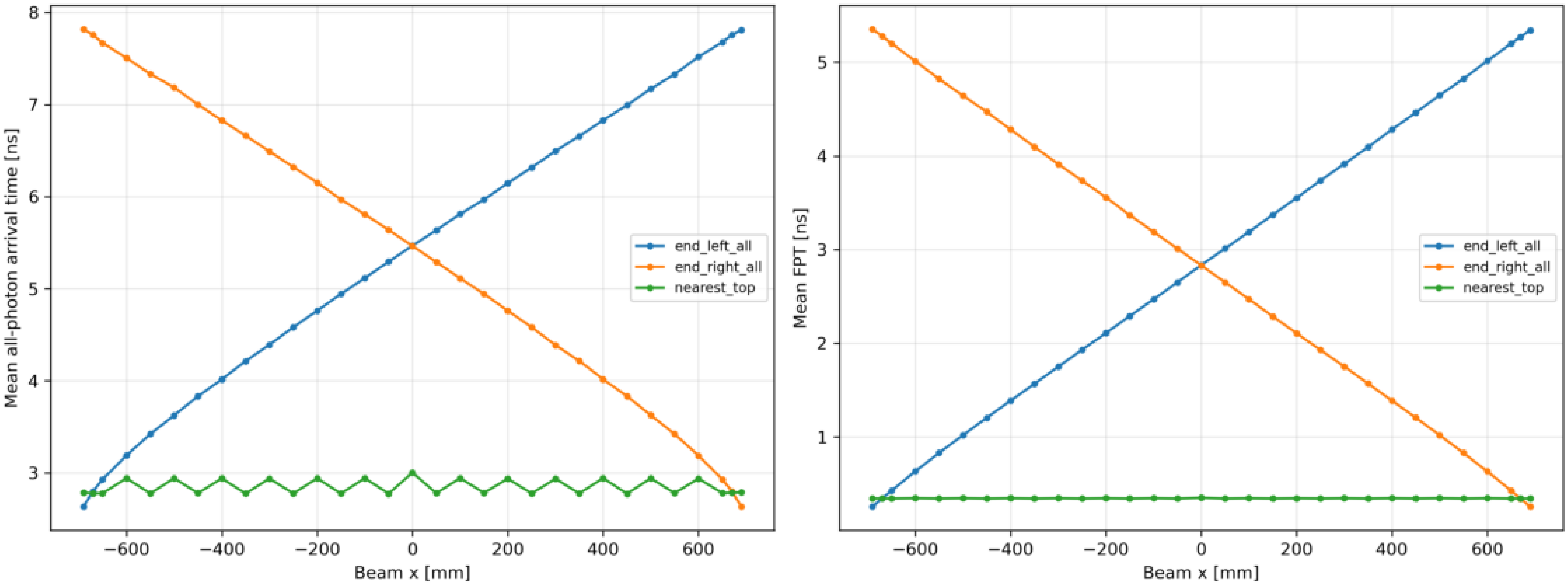


nearest_top
end_left_A_SUM4
end_left_B_SUM4
end_right_A_SUM4
end_right_B_SUM4
end_left_all
end_right_all
Poisson
var(N_pe) / mean(N_pe)
10^2
10^1
10^0
−600
−400
−200
0
200
400
600
Beam x [mm]


Poisson tail check, x=0 mm
nearest_top
F=23.27, chi2/ndf P=11.4, G=4.7
nearest_end_SUM4
F=6.38, chi2/ndf P=25.1, G=6.2
nearest_end_all
F=11.58, chi2/ndf P=18.6, G=5.9
Poisson
Gaussian
data

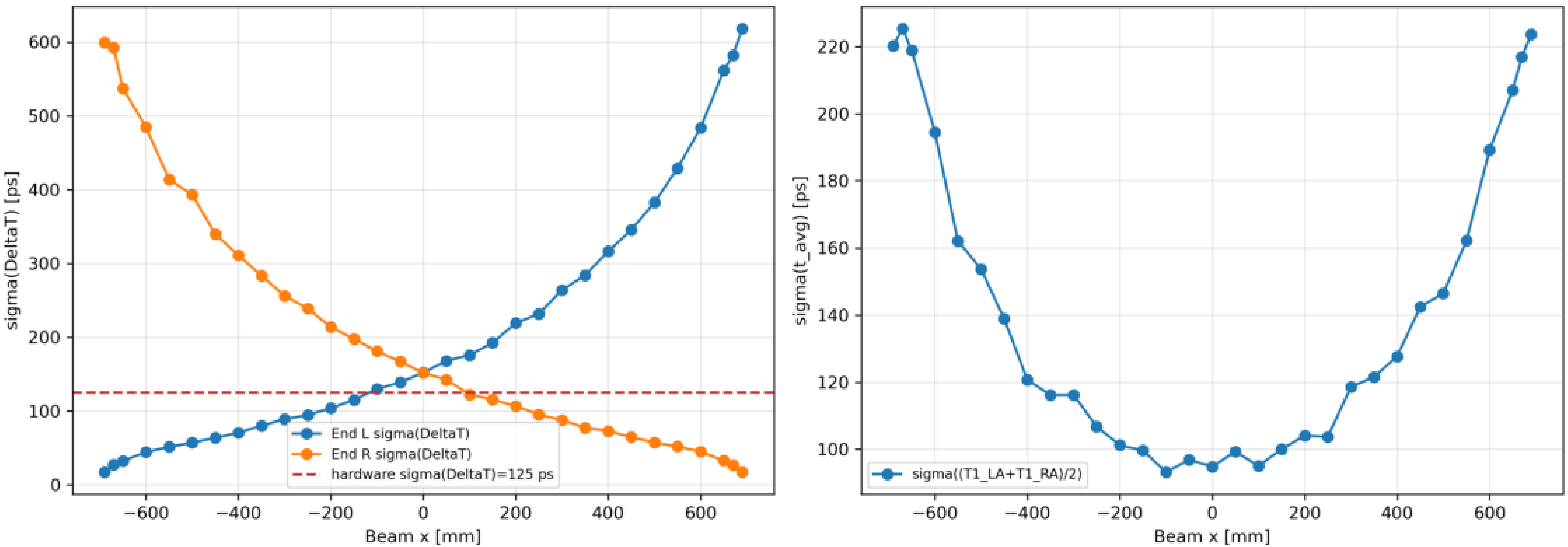


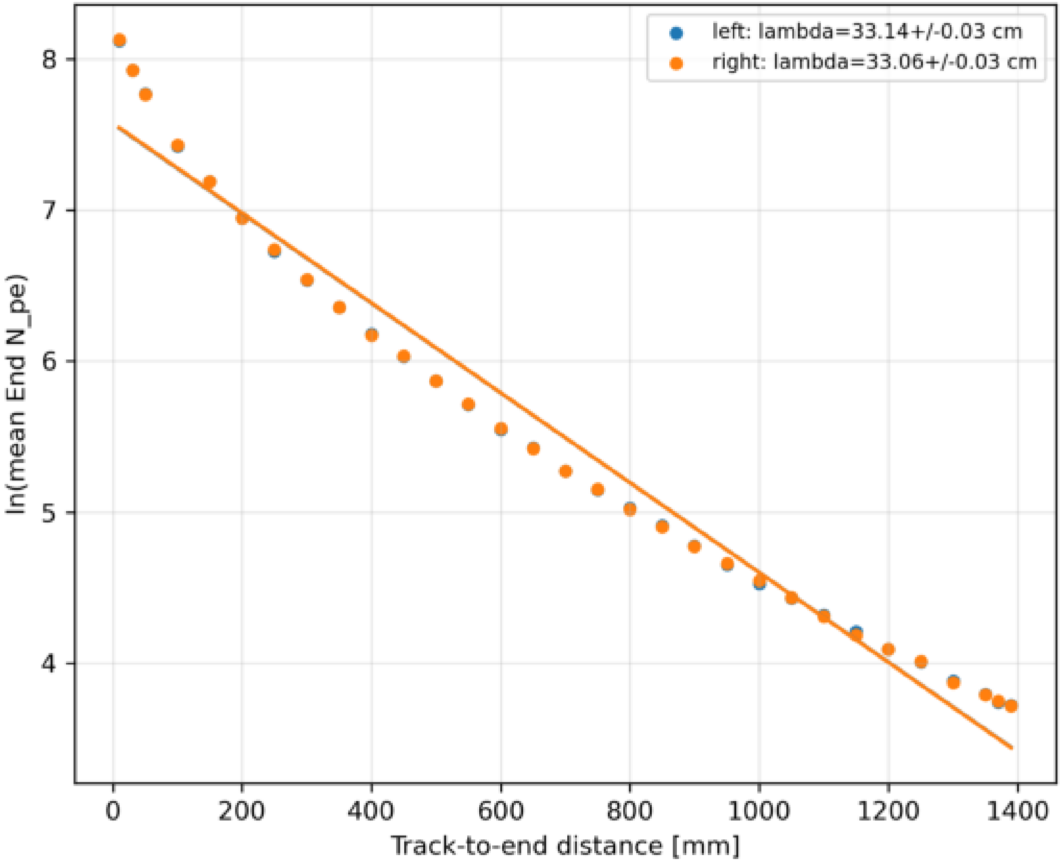
x=-690 mm, nearest_top
x=-690 mm, nearest_end_all
x=-400 mm, nearest_top
x=-400 mm, nearest_end_all
x=0 mm, nearest_top
x=0 mm, nearest_end_all
all photons
FPT
Arrival time [ns]



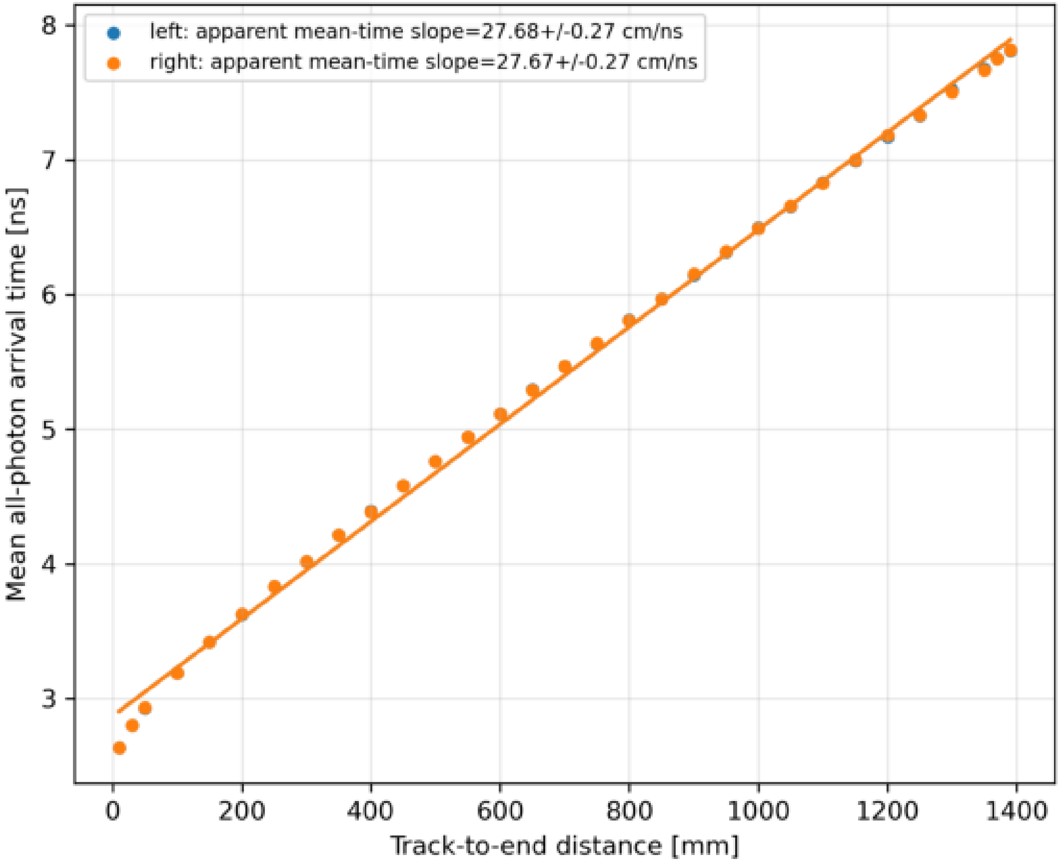
Mean all-photon arrival time [ns]
Beam x [mm]
end_left_all
end_right_all
nearest_top
Mean FPT [ns]
Beam x [mm]
end_left_all
end_right_all
nearest_top

Intrinsic End timing, no SPTR/electronics jitter





left: lambda=33.14+/-0.03 cm
right: lambda=33.06+/-0.03 cm
ln(mean End N_pe)
Track-to-end distance [mm]



left: apparent mean-time slope=27.68+/-0.27 cm/ns
right: apparent mean-time slope=27.67+/-0.27 cm/ns
Mean all-photon arrival time [ns]
Track-to-end distance [mm]